

2008 Ford Edge SE

2008 GENERAL INFORMATION Brake System - General Information - Edge & MKX

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SPECIFICATIONS

MATERIAL

Item	Specification	Fill Capacity
High Performance DOT 3 Motor Vehicle Brake Fluid PM-1-C (US); CPM-1-C (Canada)	WSS-M6C62-A or WSS-M6C65-A1	803 - 821 ml (1.69 - 1.74 pt)
High Temperature Nickel Anti-Seize Lubricant XL-2 (US); CXG-2-B (Canada)	ESE-M12A4-A	-
Metal Brake Parts Cleaner PM-4-A or PM-4-B (US); CPM-4 (Canada)	-	-
Silicone Brake Caliper Grease and Dielectric Compound XG-3-A	ESE-M1C171-A	-

GENERAL SPECIFICATIONS

Item	Specification
Front Brake Disc	
Maximum brake pad taper wear (in any direction)	3.0 mm (0.118 in)
Minimum brake disc thickness	26 mm (1.023 in)
Minimum brake pad thickness	3.0 mm (0.118 in)
Rear Brake Disc	
Maximum brake pad taper wear (in any direction)	3.0 mm (0.118 in)
Minimum brake disc thickness	16 mm (0.629 in)
Minimum brake pad thickness	3.0 mm (0.118 in)

TORQUE SPECIFICATIONS

Description	N.m	lb-ft	lb-in
Brake caliper bleeder screw	8	-	71
Master cylinder brake tube fittings	28	21	-

DESCRIPTION AND OPERATION

BRAKE SYSTEM

The brake system has the following components:

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- Fixed position foot pedal
- Vacuum-assisted power brake booster
- Diagonal split hydraulic system
- Front disc brakes
- Rear disc brakes
- Drum-in-hat park brake system
- ABS hydraulic control unit (HCU).

The brake pedal is connected to the power brake booster, which is connected to the brake master cylinder. When the brake pedal is pressed, brake fluid is pushed through the doubled-walled steel tubes and flexible hoses to the front and rear disc brake calipers. The brake fluid enters the disc brake calipers, forcing the caliper pistons and brake pads outward against the brake disc friction surface, slowing or stopping rotation. When the brake pedal is released, brake fluid pressure is relieved, returning the front and rear disc brake caliper pistons and brake pads to the unapplied position.

For information on:

- the non-adjustable pedals, refer to **HYDRAULIC BRAKE ACTUATION** article.
- ABS and HCU, refer to **VEHICLE DYNAMIC SYSTEMS** article.
- the master cylinder and fluid reservoir, refer to **HYDRAULIC BRAKE ACTUATION** article.
- the vacuum assisted power brake booster, refer to **POWER BRAKE ACTUATION** article.
- the front brake calipers, refer to **FRONT DISC BRAKE** article.
- the rear brake calipers, refer to **REAR DISC BRAKE** article.
- the drum-in-hat integrated parking brake system, refer to **PARKING BRAKE AND ACTUATION** article.

DIAGNOSTIC TESTS

BRAKE SYSTEM

MATERIAL

Item	Specification
High Performance DOT 3 Motor Vehicle Brake Fluid PM-1-C (US); CPM-1-C (Canada)	WSS-M6C62-A or WSS-M6C65-A1
Metal Brake Parts Cleaner PM-4-A or PM-4-B (US); CPM-4 (Canada)	-
Silicone Brake Caliper Grease and Dielectric Compound XG-3-A	ESE-M1C171-A

Principles of Operation

Brake System

Applying the brake pedal uses lever action to push a rod into the brake booster, which through the use of vacuum, boosts the force of the rod and then transmits this force into the master cylinder. This produces hydraulic pressure in the master cylinder. On vehicles not equipped with ABS, the hydraulic pressure is transmitted by brake fluid through the brake tubes to the individual brake calipers. On vehicles equipped with ABS, the hydraulic pressure is transmitted by brake fluid through the brake tubes to the ABS hydraulic control unit (HCU), which then distributes that pressure to the individual brake calipers. The brake calipers use hydraulic pressure to apply the pads. The application of the brake pads will cause the rotation of the wheels to slow or stop depending on how much brake pressure is applied. The parking brakes carry out the same function except that they are mechanically actuated by a cable that connects only to the rear brakes.

Red Brake Warning Indicator

The red brake warning indicator alerts the driver to certain conditions that exist in the brake system. The instrument cluster performs a bulb check when the ignition key is turned to the RUN position. The conditions that cause the indicator to illuminate are low brake fluid level, the parking brake is applied or there is a fault in the ABS (if the yellow ABS warning indicator is also illuminated).

Inspection and Verification

WARNING: Do not use any fluid other than clean brake fluid meeting manufacturer's specification. Additionally, do not use brake fluid that has been previously drained. Following these instructions will help prevent system contamination, brake component damage and the risk of serious personal injury.

WARNING: Carefully read cautionary information on product label. For EMERGENCY MEDICAL INFORMATION seek medical advice. In the USA or Canada on Ford/Motorcraft products call: 1-800-959-3673. For additional information, consult the product Material Safety Data Sheet (MSDS) if available. Failure to follow these instructions may result in serious personal injury.

NOTE: Brake fluid is harmful to painted and plastic surfaces. If brake fluid is spilled onto a painted or plastic surface, immediately wash it with water.

NOTE: Always check the fluid level in the brake master cylinder reservoir before carrying out the test procedures. If the fluid level is not at the correct level, clean the reservoir cap before removing, then add clean, specified brake fluid.

NOTE: Prior to carrying out any diagnosis, make sure the brake warning indicator operates correctly. If the brake indicator does not operate correctly, refer to **INSTRUMENT CLUSTER (IC), MESSAGE CENTER, AND WARNING CHIMES** article.

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The first indication that something may be wrong in the brake system is a change in the feeling through the brake pedal. The brake warning indicator in the instrument cluster and the brake fluid level in the brake master cylinder reservoir are also indicators of system concerns.

If a wheel is locked and the vehicle must be moved, open a bleeder screw at the locked wheel to let out enough fluid to relieve the pressure. Close the bleeder screw. This bleeding operation may release the brakes but will not correct the cause of the concern. If this does not relieve the locked wheel condition, repair the locked components before proceeding.

Inspect all hoses and connections. All unused vacuum connectors should be capped. Make sure hoses and their connections are correctly secured and in good condition with no holes, soft or collapsed areas.

1. Verify the customer concern.
2. Visually inspect for obvious signs of mechanical or electrical damage.

VISUAL INSPECTION CHART

Mechanical	Electrical
• Brake master cylinder • Brake master cylinder reservoir • Brake booster • Brake booster check valve • Brake booster vacuum hose • Brake hoses and tubes • Brake caliper • Brake caliper piston seals • Brake caliper guide pins • Brake disc • Brake pads • Brake pedal linkage • Wheel bearings • Tires	• Parking brake switch • Brake fluid level switch • Wiring, terminals or connectors

3. If an obvious cause for an observed or reported concern is found, correct the cause (if possible) before proceeding to the next step.
4. If the cause is not visually evident, go to **SYMPTOM CHART - BRAKE SYSTEM** table or Go to **SYMPTOM CHART - NOISE, VIBRATION AND HARSHNESS (NVH)** table.

For low or spongy brake pedal concerns:

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- check for fluid leaks. Repair as necessary.
- check and, if necessary, refill the brake master cylinder reservoir with clean, specified brake fluid.
- bleed the brake system and retest the brake pedal feel.
- if the brake pedal is still low or feels spongy, check the brake pedal mounting for looseness and correct installation. Check the power brake booster and the master cylinder for loose mounting. Correct as necessary and retest the system for normal operation.

For a slow or incomplete brake pedal return concern:

- inspect for binding, damage, correct installation or interference at the brake pedal.
- check the power brake booster for binding, damage and correct installation.

Road Test

The technician should have a thorough knowledge of the brake system operation and accepted general braking guidelines in order to detect any problems.

Select a road that is reasonably smooth and level. Gravel or bumpy roads are not suitable because the surface does not allow the tires to grip the road equally. Avoid crowned roads.

A key factor in evaluating brake concerns is the deceleration rate. This varies from vehicle to vehicle and with changes in operating conditions. It is evident how well the brakes are working after just a few applications.

Brake Pads

NOTE: It is not required to install new brake pads if the friction material thickness is within specifications. It is also not required to install new brake pads when the brake discs are machined.

- Remove the brake pads. Refer to **FRONT DISC BRAKE** article for front disc brakes or **REAR DISC BRAKE** article for rear disc brakes.
- Inspect and measure the thickness of the brake pad friction material. Refer to **SPECIFICATIONS**.
 - If there are missing chunks or cracks in the lining through to the backing plate, install new brake pads. Refer to **FRONT DISC BRAKE** article for front disc brakes or **REAR DISC BRAKE** article for rear disc brakes.
 - If the thickness of the friction material is less than the specified thickness, install new brake pads. Refer to **FRONT DISC BRAKE** article for front disc brakes or **REAR DISC BRAKE** article for rear disc brakes.
 - If the friction material shows taper wear that is not within specifications, verify the caliper guide pins are functioning correctly. Refer to Brake Caliper Guide Pins.

Brake Discs

NOTE: Using an impact tool without a torque socket will lead to unevenly tightened

wheel nuts. This causes brake disc on-vehicle lateral runout and brake roughness.

NOTE: It is generally not required to install new brake discs to address noise issues.

- Remove the brake disc. Refer to **FRONT DISC BRAKE** article for front disc brakes or **REAR DISC BRAKE** article for rear disc brakes.
- Inspect the brake discs and measure the brake disc thickness in a minimum of 4 places around the circumference of the brake disc. Record the measurements, refer to **SPECIFICATIONS**.
 - If the brake disc is cracked or otherwise damaged, install new a brake disc. Refer to **FRONT DISC BRAKE** article for front disc brakes or **REAR DISC BRAKE** article for rear disc brakes.
 - If any thickness measurement is less than the minimum specification, install a new brake disc. Refer to **FRONT DISC BRAKE** article for front disc brakes or **REAR DISC BRAKE** article for rear disc brakes.
 - If the diagnosis has revealed vibration in the steering wheel, seat or pedal while braking that varies with vehicle speed, machine the brake disc. Heavily scored brake discs, similar to that caused by pads worn down to the backing plate, should also be machined. In order to machine, discs must be above the minimum thickness specification. Refer to **SPECIFICATIONS** and **BRAKE DISC MACHINING**.

Brake Calipers

Inspect the brake calipers for the following:

- Brake fluid leaks
- Boots and seals for tears or cracks
- Caliper pistons for binding and corrosion
- Guide pins for correct operation. Refer to Brake Caliper Guide Pins.

Brake Caliper Guide Pins

NOTE: Do not use power tools for caliper guide pin bore cleaning. Damage to the bore may result.

NOTE: The guide pins are part of the anchor plate.

The guide pins should slide with a reasonable amount of hand force. If the brake pads show taper wear or the guide pins are difficult to move, install a new brake caliper anchor plate. Refer to **FRONT DISC BRAKE** article for the front brake caliper anchor plate or **REAR DISC BRAKE** article for the rear brake caliper anchor plate.

If the lining is not within specifications, install new brake pads. Refer to **FRONT DISC BRAKE** article for front disc brakes or **REAR DISC BRAKE** article for rear disc brakes.

Brake Hoses and Tubes

NOTE: **Never use copper tubing. It is subject to fatigue, cracking and corrosion, which may result in brake tube failure.**

- Double-wall steel tubing is used throughout the brake hydraulic system. All brake tube fittings must be correctly double flared to provide strong, leakproof connections. When bending tubing to fit the underbody or rear axle contours, be careful not to kink or crack the tube.
- If a section of the brake tube is damaged, the entire section must be installed with a new tube of the same type, size, shape and length.
- When installing the hydraulic brake tubing, hoses or connectors, tighten all connections to specifications. After installation, bleed the brake system. Refer to **BRAKE SYSTEM BLEEDING**.

Install a new brake flexible hose if the hose shows signs of softening, cracking or other damage.

When installing a new brake hose, position the hose to avoid contact with other vehicle components.

Non-Pressure Leaks

Two parts of the brake system that may have a brake fluid loss that does not appear when the system is under pressure are the brake master cylinder reservoir and the brake caliper under the following conditions:

Reservoir

- Missing or poorly-fitted brake master cylinder filler cap
- Missing or damaged brake master cylinder filler cap gasket
- Punctured or otherwise damaged brake master cylinder reservoir
- Missing, damaged or poorly-fitted sealing grommets between the brake master cylinder and the brake master cylinder reservoir

The brake master cylinder reservoir grommets are not serviceable and must be installed new as part of a new brake master cylinder and reservoir assembly. Refer to brake master cylinder in **HYDRAULIC BRAKE ACTUATION** article.

Brake Caliper

- Foreign material in the caliper piston seal groove
- Punctured or otherwise damaged caliper piston seal

The brake caliper piston seals are not serviceable and must be installed new as part of a new brake caliper. Refer to **FRONT DISC BRAKE** article for front disc brakes or **REAR DISC BRAKE** article for rear disc brakes.

Brake Master Cylinder - Normal Conditions

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The following conditions are considered normal and are not indications that the brake master cylinder is in need of service.

Condition 1: During normal operation of the brake master cylinder, the fluid level in the brake master cylinder reservoir will fall during brake application and rise during release. The net fluid level (such as after brake application and release) will remain unchanged.

Condition 2: A trace of brake fluid will exist on the booster shell below the master cylinder mounting flange. This results from the normal lubricating action of the master cylinder bore and seal.

Condition 3: Fluid level will decrease with pad wear.

Brake Master Cylinder - Abnormal Conditions

Changes in brake pedal feel or travel are indicators that something may be wrong in the brake system. Go to **SYMPTOM CHART - BRAKE SYSTEM** table for abnormal condition diagnosis.

Brake Booster

Inspect the brake booster:

- for excessive corrosion or damage.
- vacuum connections for leakage.
- vacuum hoses for kinks or leakage.
- check valve for correct operation.

Parking Brake

Check the operation of the parking brake system with the vehicle on a hoist and the parking brake control fully released. Check the rear brake adjustment or carry out the brake system diagnosis. Refer to **PARKING BRAKE AND ACTUATION** article.

Symptom Chart - Brake System

SYMPTOM CHART - BRAKE SYSTEM

Condition	Possible Sources	Action
• The red brake warning indicator and the yellow ABS warning indicator are illuminated	• DTCs in the ABS module	• REFER to <u>VEHICLE DYNAMIC SYSTEMS</u> article to diagnose the ABS module.
• The red brake warning indicator is always/never on	• Brake fluid level switch • Parking brake switch • Wiring, terminals or connectors	• REFER to <u>INSTRUMENT CLUSTER (IC), MESSAGE CENTER, AND WARNING CHIMES</u> article to diagnose the red brake warning

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	• Instrument cluster • Smart junction box (SJB)	indicator.
• The brakes pull or drift	• Tires • Brake pads • Brake discs • Brake calipers and/or guide pins • Suspension component and/or wheel alignment	• INSPECT the tires for uneven or excessive wear, and correct inflation. REFER to <u>WHEELS AND TIRES</u> article. • INSPECT the brake pads for contamination, uneven taper or excessive wear, REFER to Brake Pads. • INSPECT the brake discs and the hubs for contamination or damage. INSTALL new brake discs or hubs as necessary. REFER to <u>FRONT DISC BRAKE</u> article for front disc brakes or <u>REAR DISC BRAKE</u> article for rear disc brakes. REFER to <u>FRONT SUSPENSION</u> article for front hubs or <u>REAR SUSPENSION</u> article for rear hubs. • INSPECT the brake calipers and guide pins, REFER to Brake Calipers and Brake Caliper Guide Pins. • INSPECT the suspension and CHECK the wheel alignment. REFER to <u>SUSPENSION SYSTEM - GENERAL INFORMATION</u> article.
• Brake pedal goes down fast	• Brake fluid leaks and/or air in the system • Brake master cylinder • Hydraulic control unit (HCU)	• INSPECT the system for leaks. REPAIR as necessary. FILL the brake master cylinder reservoir. BLEED the system. REFER to <u>BRAKE SYSTEM BLEEDING</u>. • CARRY OUT the Brake Master Cylinder test in <u>COMPONENT TESTS</u>. • INSPECT the HCU. REFER to <u>VEHICLE DYNAMIC SYSTEMS</u> article.

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• The brake pedal eases down slowly	• Air in the system • Brake master cylinder • HCU	• BLEED the system. REFER to <u>BRAKE SYSTEM BLEEDING</u>. • CARRY OUT the Brake Master Cylinder test in <u>COMPONENT TESTS</u>. • INSPECT the HCU. REFER to <u>VEHICLE DYNAMIC SYSTEMS</u> article.
• Brakes lock up under light brake pedal force	• Brake pads • Brake calipers and/or guide pins • Anti-lock brake control system	• INSPECT the brake pads for uneven taper or excessive wear, REFER to Brake Pads. • INSPECT the brake calipers and guide pins, REFER to Brake Calipers and Brake Caliper Guide Pins. • CHECK the anti-lock brake control system. REFER to <u>VEHICLE DYNAMIC SYSTEMS</u> article.
• Excessive/erratic brake pedal travel	• Brake fluid leaks and/or air in the system • Brake calipers and/or guide pins • Brake flexible hose • Brake master cylinder • Brake pads	• INSPECT the system for leaks. REPAIR as necessary. BLEED the system. REFER to <u>BRAKE SYSTEM BLEEDING</u>. • INSPECT the brake calipers and guide pins, REFER to Brake Calipers and Brake Caliper Guide Pins. • INSPECT the brake flexible hoses for swelling and damage. INSTALL new as necessary. REFER to <u>FRONT DISC BRAKE</u> article for front disc brakes or <u>REAR DISC BRAKE</u> article for rear disc brakes. • CARRY OUT the Brake Master Cylinder test in <u>COMPONENT TESTS</u>. • INSPECT the brake pads for taper wear, REFER to Brake Pads.

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	• Brake pedal • Anti-lock brake control system	• INSPECT the brake pedal for binding, obstructions and correct installation. REPAIR as necessary. CHECK the brake pedal fasteners for correct torque. REFER to Specifications in <u>HYDRAULIC BRAKE ACTUATION</u> article. • CHECK the anti-lock brake control system. REFER to <u>VEHICLE DYNAMIC SYSTEMS</u> article.
• Brakes drag	• Parking brake component • Brake caliper guide pins • Disc brake caliper(s) • Brake booster • Brake master cylinder	• REPAIR or INSTALL new components as necessary. REFER to <u>PARKING BRAKE AND ACTUATION</u> article. • INSPECT the caliper guide pins for correct operation. INSTALL new components as necessary. REFER to Brake Caliper Guide Pins. • INSPECT the brake calipers. REFER to Brake Calipers. REPAIR or INSTALL new brake calipers as necessary. REFER to <u>FRONT DISC BRAKE</u> article for front disc brakes or <u>REAR DISC BRAKE</u> article for rear disc brakes. • CARRY OUT the Brake Booster test in <u>COMPONENT TESTS</u>. • CARRY OUT the Brake Master Cylinder test in <u>COMPONENT TESTS</u>.
• Excessive brake pedal effort	• Insufficient engine vacuum for brake booster operation • Brake booster manifold	• CHECK engine vacuum. REFER to the Intake Manifold Vacuum Test in <u>ENGINE SYSTEM - GENERAL INFORMATION</u> article. • VERIFY engine vacuum at the

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	vacuum hose	booster. REROUTE, REPAIR or INSTALL new components as necessary.
-	• Brake booster	• VERIFY engine vacuum at the booster. CARRY OUT the Brake Booster test in <u>COMPONENT TESTS</u> .
-	• Brake booster check valve	• VERIFY engine vacuum at the booster. CARRY OUT the Check Valve test in <u>COMPONENT TESTS</u> .

Symptom Chart - Noise, Vibration and Harshness (NVH)

NOTE: Noise, vibration and harshness (NVH) symptoms should be identified using the diagnostic tools that are available. For a list of these tools, an explanation of their uses and a glossary of common terms, refer to **NOISE, VIBRATION AND HARSHNESS** article. Since it is possible any one of multiple systems may be the cause of a symptom, it may be necessary to use a process of elimination type of diagnostic approach to pinpoint the responsible system. If this is not the causal system for the symptom, refer back to **NOISE, VIBRATION AND HARSHNESS** article for the next likely system and continue diagnosis.

SYMPTOM CHART - NOISE, VIBRATION AND HARSHNESS (NVH)

Condition	Possible Sources	Action
• Vibration when the brakes are applied	• Tires	• CORRECT any wheel and tire concern before diagnosing the brake or suspension systems. REFER to <u>WHEELS AND TIRES</u> article to diagnose the tires.
-	• Uneven brake pad wear	• COMPLETE the brake system inspection described in <u>INSPECTION AND VERIFICATION</u> . If the condition still exists, go to Pinpoint Test A .
-	• Brake disc pad transfer	
-	• Uneven brake disc wear	
-	• Suspension components	
• Brake vibration/shudder - occurs when the brake pedal is released	• Brake caliper guide pins	• INSPECT the brake caliper guide pins for correct operation. REFER to Brake Caliper Guide Pins.
-	• Brake drag	• Go to <u>SYMPTOM CHART - BRAKE SYSTEM</u> table.

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• Rattling noise	• Caliper mounting bolts loose	• CHECK the caliper bolts. TIGHTEN to specifications. REFER to <u>FRONT DISC BRAKE</u> article for front disc brakes or <u>REAR DISC BRAKE</u> article for rear disc brakes.
• Clicking noise - with brakes applied on vehicles equipped with ABS brakes	• Damaged or worn caliper guide pins or retainers • Missing or damaged anti-rattle clips or springs • Loose brake disc shield • ABS hydraulic control unit (HCU)	• CHECK the caliper guide pins and retainers for correct operation. REFER to Brake Caliper Guide Pins. INSTALL new components as necessary. • CHECK the brake pads for missing clips or broken springs. INSTALL new components as necessary. REFER to <u>FRONT DISC BRAKE</u> article for front disc brakes or <u>REAR DISC BRAKE</u> article for rear disc brakes. • TIGHTEN the brake disc shield bolts to specification. REFER to <u>FRONT DISC BRAKE</u> article for front disc brakes or <u>REAR DISC BRAKE</u> article for rear disc brakes. • Acceptable condition during an ABS event.
• Squealing noise - occurs on first (morning) brake application	• Brake pads	• Acceptable condition. Caused by humidity and low brake pad temperature.
• Squealing noise - a continuous squeal	• Brake pads worn below minimum thickness	• INSPECT brake pads for excessive wear, taper wear or uneven wear. VERIFY brake pads are within minimum specifications. REFER to Brake Pads.
• Squealing noise - an intermittent squeal brought on by cold, heat, water, mud or snow	• Brake pads	• Acceptable condition.
• Groaning noise - occurs at low speeds with brake lightly applied (creeping)	• Brake pads	• Acceptable condition.

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Grinding noise - continuous	Brake pads worn below minimum thickness	INSPECT the brake pads, brake discs, and attaching hardware for damage. VERIFY brake components are within specifications. REFER to <u>SPECIFICATIONS</u>.
Moaning noise	Brake pads contaminated with grease or oil	INSPECT the brake pads for contamination. REPAIR or INSTALL new components as necessary. REFER to <u>FRONT DISC BRAKE</u> article for front disc brakes or <u>REAR DISC BRAKE</u> article for rear disc brakes.

Pinpoint Tests

Pinpoint Test A: Vibration When Brakes are Applied

Normal Operation

During moderate to heavy braking, noise from the hydraulic control unit (HCU) and pulsation in the brake pedal can be observed. Pedal pulsation coupled with noise during heavy braking or on loose gravel, bumps, wet or snowy surfaces is acceptable and indicates correct functioning of the ABS. Pedal pulsation alone or steering wheel nibble when the brakes are applied (frequency is proportional to the vehicle speed) indicates a concern with a brake or suspension component.

This pinpoint test is intended to diagnose the following:

- Brake pad(s)
- Brake disc(s)
- Wheel(s) and tire(s)
- Suspension component(s)

PINPOINT TEST A: VIBRATION WHEN THE BRAKES ARE APPLIED

A1 ROAD TEST THE VEHICLE - LIGHT BRAKING

- Road test the vehicle. Warm the brakes by slowing the vehicle from 80 to 32 km/h (50 to 20 mph) using light brake force. At highway speeds of 89-97 km/h (55-60 mph), apply the brake using light pedal force.
- Is there a vibration/shudder felt in the steering wheel, seat or brake pedal?**
YES : Go to A4.
NO : Go to A2.

A2 ROAD TEST THE VEHICLE - MODERATE TO HEAVY BRAKING

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- Road test the vehicle. At highway speeds of 89-97 km/h (55-60 mph), apply the brake using a moderate to heavy pedal force.

- **Is there a vibration/shudder?**

YES : Go to A3.

NO : The concern is not present at this time.

A3 CHECK ABS OPERATION

- Road test the vehicle and apply the brakes on a dry, firm surface, then apply the brakes on a wet, snowy or loose surface (such as gravel).

- **Is the vibration/shudder only present on a wet, snowy or loose surface?**

YES : This is a normal operating condition of the ABS.

NO : Go to A4.

A4 CHECK THE FRONT SUSPENSION

- Check the front suspension. Refer to **SUSPENSION SYSTEM - GENERAL INFORMATION** article.

- **Are all the suspension components in satisfactory condition?**

YES : Go to A5.

NO : REPAIR or INSTALL new components as necessary. TEST the system for normal operation.

A5 RESURFACE THE FRONT BRAKE DISCS

NOTE: Do not use a bench lathe to machine the brake discs. A bench lathe may cause brake disc on-vehicle lateral runout and brake roughness.

NOTE: Follow the brake lathe manufacturer's instructions to machine the brake discs. After machining, make sure the brake disc meets the thickness specification.

- Resurface the front brake discs. Refer to **BRAKE DISC MACHINING**.
- Road test the vehicle.
- **Is the vibration/shudder present?**

YES : Go to A6.

NO : The concern has been repaired.

A6 CHECK THE REAR SUSPENSION

- Check the rear suspension. Refer to **SUSPENSION SYSTEM - GENERAL INFORMATION** article.

- **Are all the suspension components in satisfactory condition?**

YES : RESURFACE the rear brake discs or brake drums. REFER to **BRAKE DISC MACHINING**.

NO : REPAIR or INSTALL new components as necessary. TEST the system for normal operation.

Brake Booster

1. Disconnect the check valve from the brake booster.
2. Apply the parking brake, start the engine and place the transmission in NEUTRAL.
3. Verify that manifold vacuum is available at the check valve with the engine at idle speed and the transmission in NEUTRAL.
 - If manifold vacuum is available, stop the engine, connect the check valve and continue with Step 5.
 - If manifold vacuum is not available, continue with Step 4.
4. Disconnect the check valve from the vacuum hose and verify that manifold vacuum is available at the hose with the engine at idle speed and the transmission in NEUTRAL.
 - If manifold vacuum is available, stop the engine, install a new check valve and continue with Step 5.
 - If manifold vacuum is not available, stop the engine, connect the vacuum hose to the check valve and refer to **ENGINE SYSTEM - GENERAL INFORMATION** article to diagnose the no vacuum condition.
5. Apply the brake pedal several times to exhaust all vacuum from the system.
6. Apply the brake pedal and hold it in the applied position. Start the engine and verify that the brake pedal moves downward after the engine starts.
 - If the brake pedal moves, the brake booster is operating correctly
 - If the brake pedal does not move, install a new brake booster. Refer to **POWER BRAKE ACTUATION** article.
7. Operate the engine a minimum of 10 seconds at fast idle. Stop the engine, and let the vehicle stand for 10 minutes. Then apply the brake pedal with approximately 89 N (20 lb) of force. The brake pedal feel should be the same as that noted with the engine operating. If the brake pedal feels hard (no power assist), install a new brake booster check valve and retest. If the brake pedal feels spongy, bleed the hydraulic system to remove air. Refer to **BRAKE SYSTEM BLEEDING**.

Brake Booster Check Valve

The function of the brake booster check valve is to allow manifold vacuum to enter the brake booster and prevent the escape of vacuum in case manifold vacuum is lost during sustained full throttle operation.

1. Disconnect the vacuum booster hose from the check valve.
2. Apply the parking brake, start the engine and place the transmission in NEUTRAL.
3. Verify that manifold vacuum is available at the check valve end of the vacuum hose with the engine at idle speed and the transmission in NEUTRAL.
 - If manifold vacuum is available, stop the engine, connect the vacuum hose to the check valve and continue this test.
 - If manifold vacuum is not available, stop the engine, connect the vacuum hose to the check valve and refer to **ENGINE SYSTEM - GENERAL INFORMATION** article to diagnose the no vacuum condition.
4. Connect the vacuum hose to the check valve and run the engine for at least 10 seconds.

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5. Operate the brake pedal to check for power assist.
 - If power assist is present, continue with this test.
 - If power assist is not present, refer to Brake Booster test in **COMPONENT TESTS**.

NOTE: **Do not remove the brake booster check valve from the brake booster in this step.**

6. Stop the engine and disconnect the vacuum booster hose from the brake booster check valve.
7. Apply the brake and verify that there is enough vacuum retained in the brake booster for at least one power-assisted brake application.
 - If there is enough vacuum for at least one power-assisted brake application, the check valve is functioning correctly.
 - If there is not enough vacuum for at least one power-assisted brake application, continue with this test.
8. Inspect the brake booster for any signs of damage that may cause a leak.
 - If any damage is found, install a new brake booster and repeat this test.
 - If no damage is found, install a new check valve.

Brake Master Cylinder - Bypass Condition

1. Disconnect the brake tubes from the master cylinder.
2. Plug the outlet ports of the master cylinder.
3. Apply the brakes. If brake pedal height cannot be maintained, the brake master cylinder has an internal leak and a new brake master cylinder must be installed.

Brake Master Cylinder - Compensator Port

The purpose of the compensator ports in the brake master cylinder is to:

- supply additional brake fluid from the brake master cylinder reservoir needed by the brake system due to brake lining wear.
- allow brake fluid to return to the brake master cylinder reservoir when the brakes are released. The returning brake fluid creates a slight turbulence in the brake master cylinder reservoir. This is a normal condition and indicates that the compensator ports are not clogged.

Clogged compensator ports may cause the brakes to hang up or not fully release. If clogged compensator ports are suspected, proceed as follows:

1. With the vehicle in NEUTRAL, position it on a hoist. Refer to **JACKING AND LIFTING** article.
2. With the brakes released, attempt to rotate each wheel and check for any brake drag.
 - If an excessive amount of brake drag exists at all 4 wheels, continue with the test.
 - If an excessive amount of brake drag exists at only one wheel, it indicates a possible seized brake caliper, brake wheel cylinder or parking brake component. Repair or install new components as

necessary.

3. Check the brake stoplamp switch and the brake pedal free play to verify that the brake pedal is not partially applied.
4. Loosen the brake master cylinder nuts and position the brake master cylinder away from the brake booster.
5. With the brakes released, attempt to rotate each wheel and check for any brake drag.
 - If the brake drag is no longer present, install a new brake booster. Refer to **POWER BRAKE ACTUATION** article.
 - If the brake drag is still present, install a new master cylinder. Refer to **HYDRAULIC BRAKE ACTUATION** article.

GENERAL PROCEDURES

BRAKE DISC MACHINING

NOTE: Do not use a bench lathe to machine the brake discs. Use an on-vehicle brake lathe only. Read the entire operating manual and/or view the video shipped with the lathe before installing, operating or repairing the lathe.

NOTE: An on-vehicle brake lathe with an automatic runout adjustment feature is preferred. However, if the lathe is not self adjusting, the lathe oscillation must be adjusted using a dial indicator. The total indicated runout target is 0.000 mm (0.000 in). The maximum indicated runout should be no more than 0.050 mm (0.002 in). If the runout adjustment (automatic or manual) is carried out correctly prior to machining, then the final brake disc runout will be within specification and a runout measurement is not necessary after machining.

NOTE: Do not machine new brake discs.

NOTE: Lateral runout and disc thickness variation measurements are not required because correct adjustment of the on-vehicle brake lathe will make sure that these dimensions are within specification.

1. Remove the wheel and tire. For additional information, refer to **WHEELS AND TIRES** article.

NOTE: Do not allow the caliper to hang from the brake hose or damage to the hose may occur.

NOTE: It is not necessary to disconnect the brake tube from the brake caliper.

2. Remove the bolts and position the brake caliper or brake caliper and anchor plate assembly aside, as required.
 - Support the brake caliper using mechanic's wire.

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3. Install the hub adapter using:
 - four wheel nuts on a 4-stud wheel hub.
 - five wheel nuts on a 5-stud wheel hub.
 - six wheel nuts on a 6-stud wheel hub.
 - four wheel nuts on a 7- or 8-stud wheel hub.
 - five wheel nuts on a 10-stud wheel hub.
4. Install the cutting lathe.
5. If the lathe is not self adjusting, adjust the lathe oscillation using a dial indicator. The total indicated runout target is 0.000 mm (0.000 in). The maximum indicated runout should be no more than 0.050 mm (0.002 in).
6. Center the cutting head, adjust the cutting bits and install the chip deflector/silencer.

NOTE: The depth of the cut should be between 0.10 and 0.40 mm (0.004 and 0.015 in). Lighter cuts will cause the bit to heat up and wear faster. Heavier cuts will cause poor brake disc surface finish.

7. Machine the brake disc.
8. Remove the lathe and the silencer.
9. Remove the wheel nuts and hub adapter.
10. Remove the metal shavings.

NOTE: It is not required to install new brake pads if friction material is within specifications. For additional information, refer to SPECIFICATIONS.

11. Position the brake caliper or brake caliper and anchor plate assembly.
 - Install the bolts.
 - For fastener torque specifications, refer to FRONT DISC BRAKE article for front disc brakes or REAR DISC BRAKE article for rear disc brakes.
12. Install the wheel and tire. For additional information, refer to WHEELS AND TIRES article.

BRAKE SYSTEM BLEEDING

SPECIAL TOOLS

Illustration	Tool Name	Tool Number
 ST1137-A	Vehicle Communication Module (VCM) and Integrated Diagnostic System (IDS) software with appropriate hardware, or equivalent scan tool	-

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MATERIAL

Item	Specification
High Performance DOT 3 Motor Vehicle Brake Fluid PM-1-C (US); CPM-1-C (Canada)	WSS-M6C62-A or WSS-M6C65-A1

Pressure Bleed

WARNING: Do not use any fluid other than clean brake fluid meeting manufacturer's specification. Additionally, do not use brake fluid that has been previously drained. Following these instructions will help prevent system contamination, brake component damage and the risk of serious personal injury.

WARNING: Carefully read cautionary information on product label. For EMERGENCY MEDICAL INFORMATION seek medical advice. In the USA or Canada on Ford/Motorcraft products call: 1-800-959-3673. For additional information, consult the product Material Safety Data Sheet (MSDS) if available. Failure to follow these instructions may result in serious personal injury.

WARNING: Do not allow the brake master cylinder to run dry during the bleeding operation. Master cylinder may be damaged if operated without fluid, resulting in degraded braking performance. Failure to follow this instruction may result in serious personal injury.

NOTE: Brake fluid is harmful to painted and plastic surfaces. If brake fluid is spilled onto a painted or plastic surface, immediately wash it with water.

NOTE: When any part of the hydraulic system has been disconnected for repair or installation of new components, air can get into the system and cause spongy brake pedal action. This requires bleeding of the hydraulic system after it has been correctly connected.

NOTE: Pressure bleed the brake system at 207-345 kPa (30-50 psi).

NOTE: If the hydraulic control unit (HCU) or any component upstream of the HCU are installed new, carry out the Brake System Pressure Bleeding procedure, then the anti-lock HCU bleed procedure, followed by another Brake System Bleed Procedure.

1. Clean all the dirt from around the brake fluid reservoir cap and remove the cap. Fill the brake master cylinder reservoir with clean, specified brake fluid.

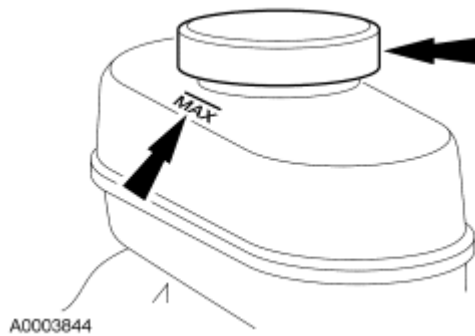


Fig. 1: Identifying Brake Master Cylinder Fluid Reservoir Cap And Max Fill Line
Courtesy of FORD MOTOR CO.

NOTE: Master cylinder pressure bleeder adapter tools are available from various manufacturers of pressure bleeding equipment. Follow the instructions of the manufacturer when installing the adapter.

2. Install the bleeder adapter to the brake master cylinder reservoir and attach the bleeder tank hose to the fitting on the adapter.

NOTE: Bleed the longest line first. Make sure the bleeder tank contains enough specified brake fluid to complete the bleeding operation.

3. Remove the RH rear brake caliper bleeder cap and place a box-end wrench on the bleeder screw. Attach a rubber drain tube to the RH rear bleeder screw and submerge the free end of the tube in a container partially filled with clean, specified brake fluid.
4. Open the valve on the bleeder tank.
5. Loosen the RH rear brake caliper bleeder screw. Leave open until clear, bubble-free brake fluid flows, then tighten the RH rear bleeder screw.
 - Press and release the parking brake 5 times.
 - Repeat until clear, bubble-free fluid comes out.
6. Tighten the RH rear brake caliper bleeder screw to 8 N.m (71 lb-in), remove the rubber hose and install the bleeder screw cap.
7. Repeat Steps 5 and 6 for the LH rear brake caliper.
 - Tighten the LH rear brake caliper bleeder screw to 8 N.m (71 lb-in).
8. Continue bleeding the front of the system, going in order from the RH front brake caliper and then to the LH front brake caliper.
 - Tighten the front brake caliper bleeder screws to 8 N.m (71 lb-in).
9. Close the bleeder tank valve. Remove the tank hose from the adapter and remove the adapter. Fill the reservoir with clean, specified brake fluid and install the reservoir cap.

Manual Bleeding

WARNING: Do not use any fluid other than clean brake fluid meeting manufacturer's

specification. Additionally, do not use brake fluid that has been previously drained. Following these instructions will help prevent system contamination, brake component damage and the risk of serious personal injury.

WARNING: Carefully read cautionary information on product label. For **EMERGENCY MEDICAL INFORMATION** seek medical advice. In the USA or Canada on Ford/Motorcraft products call: 1-800-959-3673. For additional information, consult the product Material Safety Data Sheet (MSDS) if available. Failure to follow these instructions may result in serious personal injury.

WARNING: Do not allow the brake master cylinder to run dry during the bleeding operation. Master cylinder may be damaged if operated without fluid, resulting in degraded braking performance. Failure to follow this instruction may result in serious personal injury.

NOTE: Brake fluid is harmful to painted and plastic surfaces. If brake fluid is spilled onto a painted or plastic surface, immediately wash it with water.

NOTE: When any part of the hydraulic system is disconnected for repair or installation of new components, air can get into the system and cause spongy brake pedal action. This requires bleeding of the hydraulic system after it is correctly connected. The hydraulic system can be bled manually or with pressure bleeding equipment.

1. Clean all dirt from and remove the brake master cylinder filler cap and fill the brake master cylinder reservoir with clean, specified brake fluid.

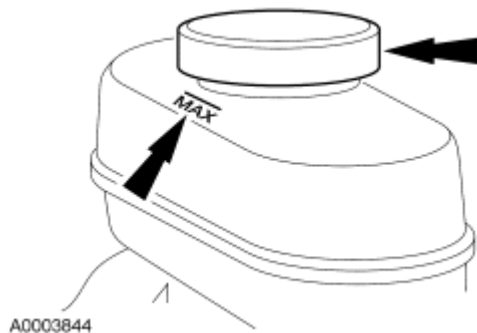


Fig. 2: Identifying Brake Master Cylinder Fluid Reservoir Cap And Max Fill Line
Courtesy of FORD MOTOR CO.

NOTE: Bleed the brake system from the longest to the shortest brake line.

2. Remove the RH rear brake caliper bleeder screw cap and place a box-end wrench on the bleeder screw. Attach a rubber drain hose to the bleeder screw and submerge the free end of the hose in a container

partially filled with clean, specified brake fluid.

3. Have an assistant pump the brake pedal and then hold firm pressure on the brake pedal.
4. Loosen the RH rear brake caliper bleeder screw until a stream of brake fluid comes out. While the assistant maintains pressure on the brake pedal, tighten the bleeder screw.
 - Repeat until clear, bubble-free fluid comes out.
 - Refill the brake master cylinder reservoir as necessary.
5. Tighten the RH rear brake caliper bleeder screw to 8 N.m (71 lb-in), remove the rubber hose and install the bleeder screw cap.
6. Repeat Steps 2 through 5 for the LH rear brake caliper bleeder screw.
 - Tighten the LH rear brake caliper bleeder screw to 8 N.m (71 lb-in).
7. Remove the RH front brake caliper bleeder screw cap and place a box-end wrench on the bleeder screw. Attach a rubber drain hose to the bleeder screw and submerge the free end of the hose in a container partially filled with clean, specified brake fluid.

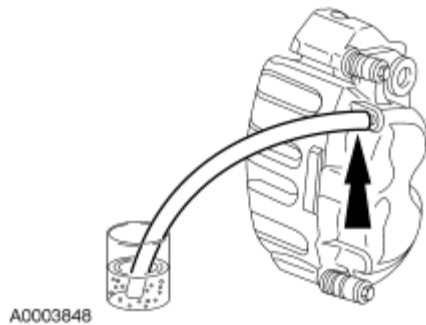


Fig. 3: Locating Rubber Drain Tube
Courtesy of FORD MOTOR CO.

8. Have an assistant pump the brake pedal and then hold firm pressure on the brake pedal.
9. Loosen the RH front brake caliper bleeder screw until a stream of brake fluid comes out. While the assistant maintains pressure on the brake pedal, tighten the bleeder screw.
 - Repeat until clear, bubble-free fluid comes out.
 - Refill the brake master cylinder reservoir as necessary.
10. Tighten the RH front brake caliper bleeder screw to 8 N.m (71 lb-in), remove the rubber hose and install the bleeder screw cap.
11. Repeat Steps 7 through 10 for the LH front brake caliper bleeder screw.
 - Tighten the LH front brake caliper bleeder screw to 8 N.m (71 lb-in).

ABS Hydraulic Control Unit (HCU) Bleeding

WARNING: Do not use any fluid other than clean brake fluid meeting manufacturer's specification. Additionally, do not use brake fluid that has been previously drained. Following these instructions will help prevent system contamination, brake component damage and the risk of serious personal

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injury.

WARNING: Carefully read cautionary information on product label. For EMERGENCY MEDICAL INFORMATION seek medical advice. In the USA or Canada on Ford/Motorcraft products call: 1-800-959-3673. For additional information, consult the product Material Safety Data Sheet (MSDS) if available. Failure to follow these instructions may result in serious personal injury.

WARNING: Do not allow the brake master cylinder to run dry during the bleeding operation. Master cylinder may be damaged if operated without fluid, resulting in degraded braking performance. Failure to follow this instruction may result in serious personal injury.

NOTE: Brake fluid is harmful to painted and plastic surfaces. If brake fluid is spilled onto a painted or plastic surface, immediately wash it with water.

NOTE: This procedure is only required when a new hydraulic control unit (HCU) is installed.

NOTE: When any part of the hydraulic system is disconnected for repair or installation of new components, air can get into the system and cause spongy brake pedal action. This requires bleeding of the hydraulic system after it is correctly connected. The hydraulic system can be bled manually or with pressure bleeding equipment.

NOTE: Carrying out the System Bleed function drives trapped air from the HCU. Subsequent bleeding removes the air from the brake hydraulic system through the bleeder screws.

NOTE: Adequate voltage to the HCU module is required during the anti-lock control portion of the system bleed.

1. Connect the scan tool to the vehicle.
2. Access the ABS HCU bleed function and follow the directions on the scan tool.
3. Manually bleed the brake hydraulic system. For additional information, refer to **MANUAL BLEEDING**.
4. Repeat the procedure carrying out a total of 2 scan tool cycles and 2 manual bleed cycles.

COMPONENT BLEEDING

MATERIAL

Item	Specification
High Performance DOT 3 Motor Vehicle Brake Fluid	WSS-M6C62-A or WSS-M6C65-A1

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PM-1-C (US); CPM-1-C (Canada)

Master Cylinder

WARNING: Do not use any fluid other than clean brake fluid meeting manufacturer's specification. Additionally, do not use brake fluid that has been previously drained. Following these instructions will help prevent system contamination, brake component damage and the risk of serious personal injury.

WARNING: Carefully read cautionary information on product label. For EMERGENCY MEDICAL INFORMATION seek medical advice. In the USA or Canada on Ford/Motorcraft products call: 1-800-959-3673. For additional information, consult the product Material Safety Data Sheet (MSDS) if available. Failure to follow these instructions may result in serious personal injury.

WARNING: Do not allow the brake master cylinder to run dry during the bleeding operation. Master cylinder may be damaged if operated without fluid, resulting in degraded braking performance. Failure to follow this instruction may result in serious personal injury.

NOTE: Brake fluid is harmful to painted and plastic surfaces. If brake fluid is spilled onto a painted or plastic surface, immediately wash it with water.

NOTE: When a new brake master cylinder has been installed or the system has been emptied, or partially emptied, it should be primed to prevent air from getting into the system.

NOTE: When any part of the hydraulic system has been disconnected for repair or new installation, air may get into the system and cause spongy brake pedal action. This requires bleeding of the hydraulic system after it has been correctly connected. The hydraulic system can be bled manually or with pressure bleeding equipment.

1. Disconnect the brake master cylinder outlet tubes.

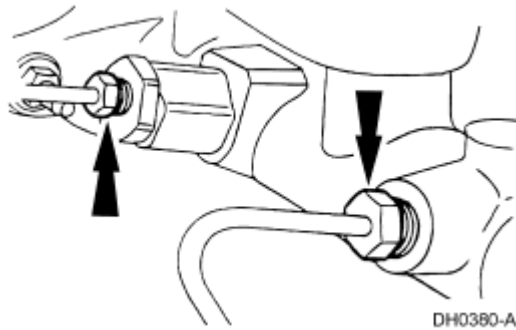


Fig. 4: Identifying Brake Lines
Courtesy of FORD MOTOR CO.

2. Install short brake tubes with ends submerged in the brake master cylinder reservoir and fill the brake master cylinder reservoir with clean, specified brake fluid.



Fig. 5: Identifying Short Brake Tubes
Courtesy of FORD MOTOR CO.

3. Have an assistant pump the brake pedal until clear fluid flows from both brake tubes without air bubbles.
4. Remove the short brake tubes and install the brake outlet tubes.
5. Bleed each brake tube at the brake master cylinder as follows:
 1. Have an assistant pump the brake pedal and then hold firm pressure on the brake pedal.
 2. Loosen the rearmost brake tube fittings until a stream of brake fluid comes out. Have an assistant maintain pressure on the brake pedal while tightening the brake tube fitting.
 3. Repeat this operation until clear, bubble-free fluid comes out.
 4. Refill the brake master cylinder reservoir as necessary. Repeat the bleeding operation at the front brake tube.
6. While the assistant maintains pressure on the brake pedal, tighten the brake tubes.
 - Tighten to 28 N.m (21 lb-ft).

Brake Caliper

WARNING: Do not use any fluid other than clean brake fluid meeting manufacturer's specification. Additionally, do not use brake fluid that has been

previously drained. Following these instructions will help prevent system contamination, brake component damage and the risk of serious personal injury.

WARNING: Carefully read cautionary information on product label. For **EMERGENCY MEDICAL INFORMATION** seek medical advice. In the USA or Canada on Ford/Motorcraft products call: 1-800-959-3673. For additional information, consult the product Material Safety Data Sheet (MSDS) if available. Failure to follow these instructions may result in serious personal injury.

WARNING: Do not allow the brake master cylinder to run dry during the bleeding operation. Master cylinder may be damaged if operated without fluid, resulting in degraded braking performance. Failure to follow this instruction may result in serious personal injury.

NOTE: Brake fluid is harmful to painted and plastic surfaces. If spilled, wipe up immediately before damage to the painted or plastic surface occurs.

NOTE: It is not necessary to do a complete brake system bleed if only the brake caliper was disconnected or installed new.

1. Remove the brake caliper bleeder screw cap and place a box-end wrench on the bleeder screw. Attach a rubber drain hose to the bleeder screw and submerge the free end of the hose in a container partially filled with clean, specified brake fluid.

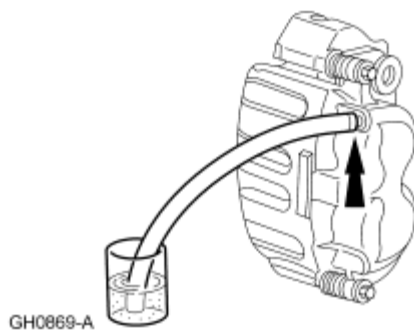


Fig. 6: Attaching Rubber Drain Hose To Disc Brake Caliper Bleeder Screw
Courtesy of FORD MOTOR CO.

2. Have an assistant pump the brake pedal and then hold firm pressure on the brake pedal.
3. Loosen the bleeder screw until a stream of brake fluid comes out. While the assistant maintains pressure on the brake pedal, tighten the bleeder screw.
 - Repeat until clear, bubble-free fluid comes out.
 - Refill the brake master cylinder reservoir as necessary.
4. Tighten the brake caliper bleeder screw, remove the rubber hose and install the bleeder screw cap.

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- Tighten the bleeder screw to 8 N.m (71 lb-in).